THE CONSTRUCTION OF THE COMPOSITION OF RELATIVE CYCLES IS COMPATIBLE WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Base change compatibility of the composition action of relative cycles (relative-cycles chapter, Lemma lemma-action-base-change, canonical tag 0H68). cited at proof lines 2834-2836. Applied on the fibres to (Y_s -> Y)^*alpha, beta_s, X_s -> Y_s -> s, and base change by s' -> s, closing the fibrewise identification.

Source reference. Original lemma 0H68: sources/stacks-chow/tags.txt, source lines 21413–21413. Cited at proof lines 2834-2836: fibrewise reduction of the assertion.

Lemma 0.2. *The formation of $\alpha \circ \beta$ is compatible with base change.*

Proof. Let $g : S' \to S$ be a morphism of schemes. Denote $X' \to Y'$ the base change of $X \to Y$ by g. Denote α' the base change of α with respect to $Y' \to Y$. Denote β' the base change of β with respect to $S' \to S$. The assertion means that $\alpha' \circ \beta'$ is the base change of $\alpha \circ \beta$ by $g : S' \to S$.

Let $s' \in S'$ be a point with image $s \in S$. Then

$$(\alpha' \circ \beta')_{s'} = (Y'_{s'} \to Y')^* \alpha' \cap \beta'_{s'}$$

We observe that

$$(Y'_{s'} \to Y')^* \alpha' = (Y'_{s'} \to Y')^* (Y' \to Y)^* \alpha = (Y'_{s'} \to Y_s)^* (Y_s \to Y)^* \alpha$$

and that $\beta'_{s'}$ is the base change of β_s by $s' = \mathrm{Spec}(\kappa(s')) \to \mathrm{Spec}(\kappa(s)) = s$. Hence the result follows from Lemma 0H68 applied to $(Y_s \to Y)^* \alpha$, β_s, $X_s \to Y_s \to s$, and base change by $s' \to s$. □

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